

Model comparison

Francisco Rodríguez-Sánchez

<https://frodriguezsanchez.net>

Trees dataset

```
trees <- read.csv('data/trees.csv')  
head(trees)
```

	site	dbh	height	sex	dead
1	4	29.68	36.1	male	0
2	5	33.29	42.3	male	0
3	2	28.03	41.9	female	0
4	5	39.86	46.5	female	0
5	1	47.94	43.9	female	0
6	1	10.82	26.2	male	0

Four models

```
m1 <- lm(height ~ dbh, data = trees)
```

```
m2 <- lm(height ~ sex, data = trees)
```

```
m3 <- lm(height ~ site, data = trees)
```

```
m4 <- lm(height ~ site*dbh, data = trees)
```

Compare model performance

```
library('performance')  
compare_performance(m1, m2, m3, m4)
```

Comparison of Model Performance Indices

Name	Model	AIC (weights)	AICc (weights)	BIC (weights)	R2
m1	lm	5660.3 (<.001)	5660.3 (<.001)	5675.0 (<.001)	0.787
m2	lm	7206.1 (<.001)	7206.2 (<.001)	7220.9 (<.001)	0.002
m3	lm	7117.3 (<.001)	7117.5 (<.001)	7171.2 (<.001)	0.102
m4	lm	5084.3 (>.999)	5085.2 (>.999)	5187.3 (>.999)	0.885

Name	R2 (adj.)	RMSE	Sigma
m1	0.787	4.089	4.093
m2	0.001	8.856	8.865
m3	0.093	8.404	8.446
m4	0.882	3.011	3.041

Compare model performance

```
library('see')  
plot(compare_performance(m1, m2, m3, m4))
```

Comparison of Model Indices



Compare parameters

```
library('parameters')  
compare_parameters(m3, m4)
```

Parameter	m3	m4
(Intercept)	33.84 (33.00, 34.68)	16.36 (15.65, 17.07)
site [2]	6.34 (4.94, 7.74)	7.68 (6.49, 8.88)
site [3]	5.00 (3.07, 6.93)	4.52 (2.82, 6.22)
site [4]	0.53 (-1.40, 2.47)	2.77 (1.17, 4.37)
site [5]	4.37 (2.52, 6.22)	3.92 (2.21, 5.63)
site [6]	4.76 (2.46, 7.06)	4.16 (2.17, 6.14)
site [7]	-0.74 (-4.37, 2.89)	-2.31 (-5.35, 0.74)
site [8]	-0.68 (-5.54, 4.17)	-2.62 (-10.64, 5.41)
site [9]	9.17 (3.25, 15.09)	2.62 (-7.34, 12.58)
site [10]	-0.58 (-8.04, 6.88)	4.66 (-1.21, 10.53)
site [3] × dbh		-6.03e-03 (-0.06, 0.05)
site [4] × dbh		-0.03 (-0.09, 0.02)
dbh		0.63 (0.61, 0.65)
site [2] × dbh		-0.04 (-0.08, 0.00)
site [7] × dbh		0.08 (-0.02, 0.18)
site [8] × dbh		-0.08 (-0.30, 0.14)
site [5] × dbh		-0.01 (-0.06, 0.04)
site [6] × dbh		1.34e-03 (-0.06, 0.06)
site [9] × dbh		0.08 (-0.21, 0.37)
site [10] × dbh		-0.10 (-0.33, 0.12)

Compare parameters

```
library('parameters')  
plot(compare_parameters(m3, m4))
```

